

---

---

# Silicon Photonics

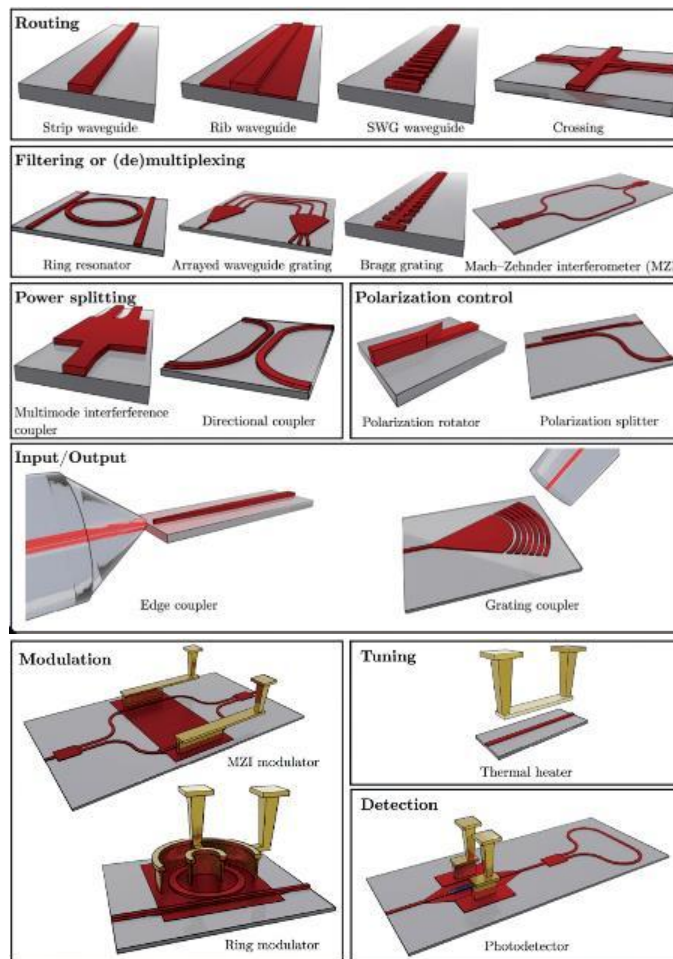
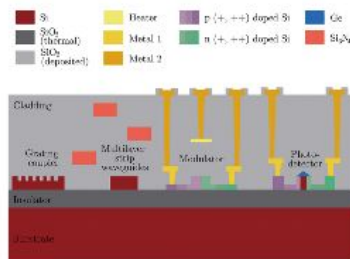
Jim

---

---

# Outline

- 矽光子介紹
- 光波導 Waveguide
- 耦合器 Directional coupler
- 環形共振器 Ring resonator
- 饋入與輸出口 I/O couple
- 干涉調變器 MRM & MZM modulator
- 鍺光偵測器 Ge on Si Photonics
- 光通訊協議 Protocol to enhance the data capacity
- 可程式光學IC Programmable PIC
- 應用場景與供應鏈



Passive component

Active component

# What is silicon photonics

Silicon photonics is a technology that constructs large-scale photonic circuits on silicon wafers using silicon semiconductor manufacturing techniques.

This technology leverages the properties of light to achieve high-speed, high-capacity, and low-power data transmission.

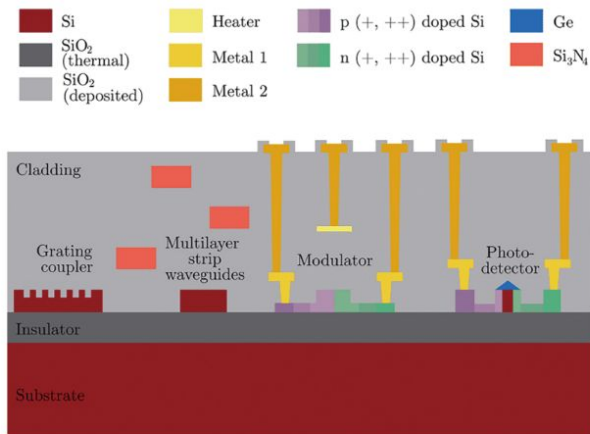
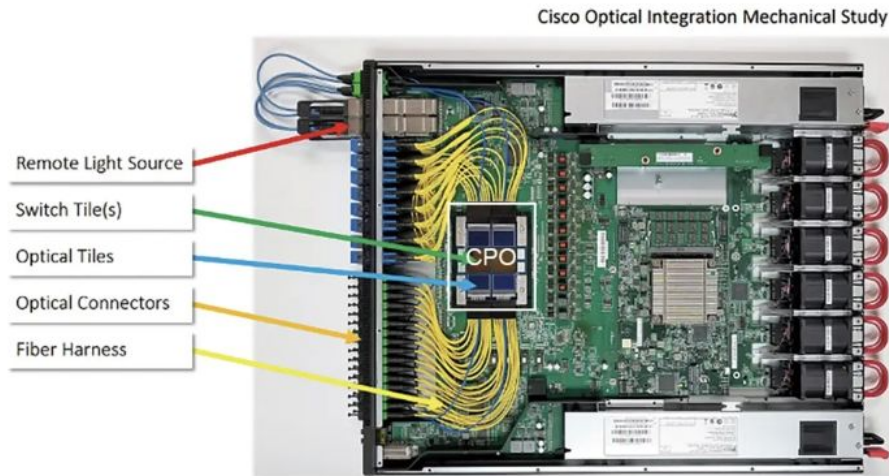


圖3 矽光子SOI晶圓剖面示意圖，紅色基板為矽，灰色絕緣層可作為下蓋光層，中間層為各式光元件，上層淺灰色為上蓋光層，黃色為導電金屬電極 (Ref. Wei Shi, et al. Nanophotonics)。



Done in collaboration with Accton

使用共同封裝光學CPO製作的光交換機。圖 / 思科 (CISCO)

# Problems Solved by Silicon photonic

**High data rate and Low Latency:** light polarization and mode , high group velocity

**Energy Efficiency:** low loss

**Scalability:** heat, crosstalk, mature silicon wafer processes(low cost)

# Why silicon photonics can achieve ?

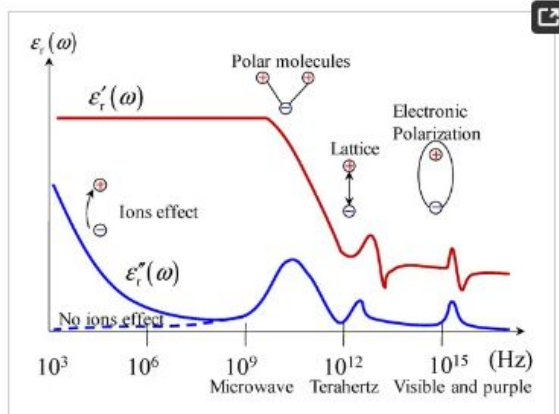


Figure 1. An illustration of frequency dependent dielectric effect.

The permittivity is actually a complex quantity. The imaginary part is due to dielectric loss. One representative mechanism of loss is the dipole relaxation [1]. When a dipole tries to follow the alternating electric field, friction between dipoles would cause loss of energy. Other mechanisms include conductive loss and lattice loss. These mechanisms can be incorporated into a single parameter the complex permittivity

$$\epsilon_r = \epsilon'_r - j\epsilon''_r \quad (1)$$

where the real part is the relative permittivity and the imaginary part is referred to as the loss factor. Since these mechanisms fall in different frequency ranges, the permittivity exhibits a dependency on frequency

$$\epsilon_r(\omega) = \epsilon'_r(\omega) - j\epsilon''_r(\omega) \quad (2)$$

Example:

FR\_4 @10<sup>10</sup>Hz : Dk 4 ,Df 0.01

SiO2 @10<sup>14</sup>Hz : Dk 1.4 ,Df 0.00001

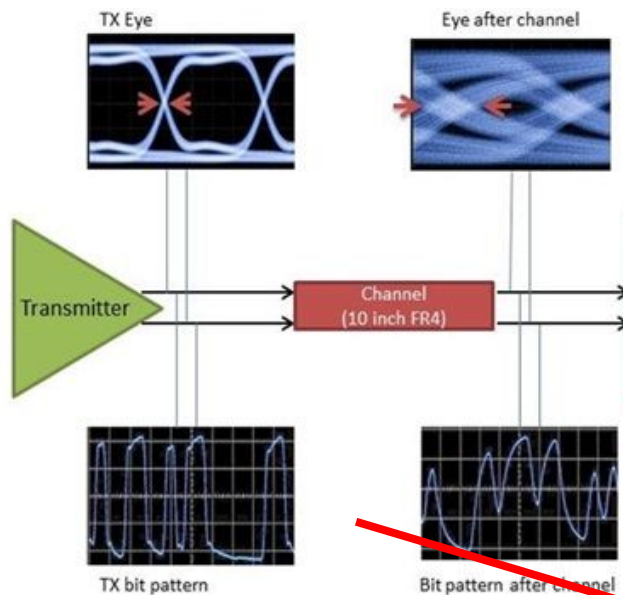
# 矽光子Q&A

## 1. 矽光子要解決的問題是什麼？

### A: 要解決目前數字信號的高速傳輸的瓶頸

信號衰減: 在長距離傳輸過程中, 信號會因為傳播損耗而逐漸衰減。這要求增設中繼站或使用放大器來維持信號的質量。

干擾和噪聲: 尤其在銅線等傳統媒介上, 外部電磁干擾和其他噪聲可以顯著影響信號質量, 導致數據錯誤和傳輸效率下降。



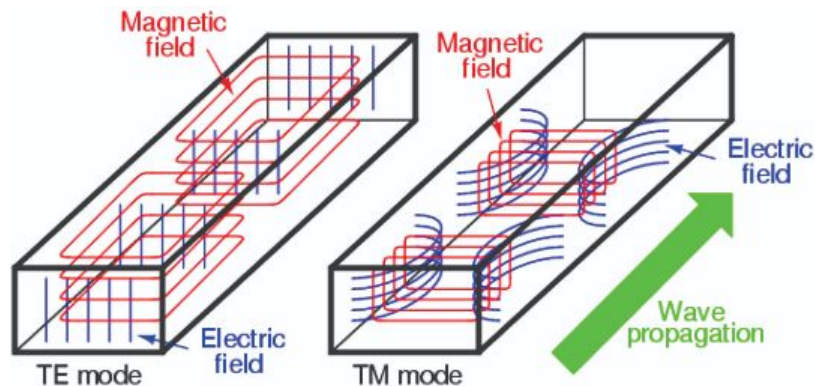
信號衰減: 光波導具有低損耗特性

干擾和噪聲: 光彼此之間不會有耦合關係

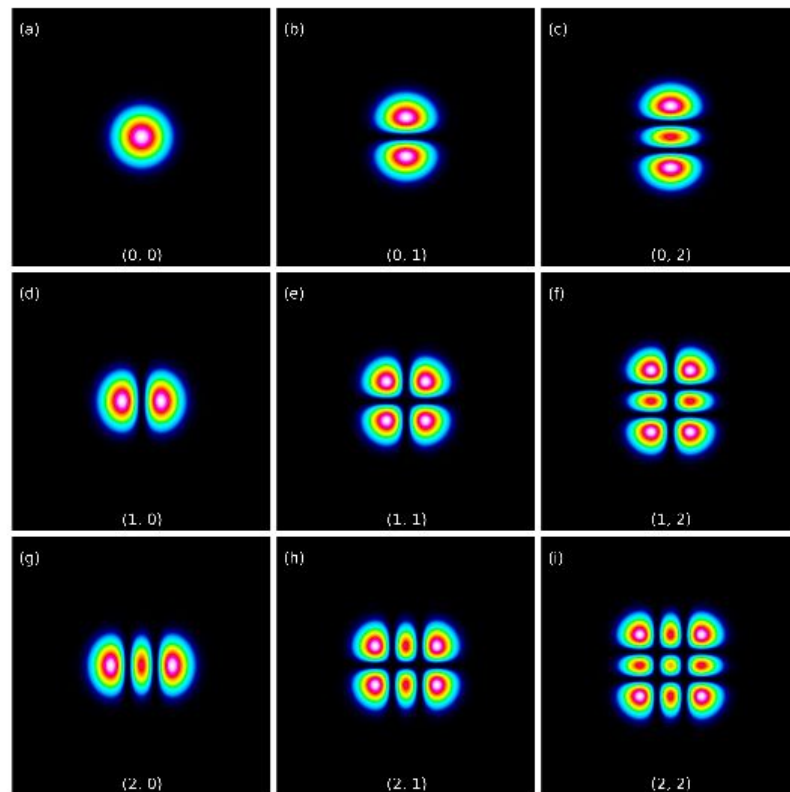
數字信號在傳輸的過程中會漸漸見失真

# Waveguide

## Principle of waveguide



- Because of the boundary condition, only certain mode can propagate in the wave guide.
- Also, we need the large difference of reflection index to get the good confinement of electric field.

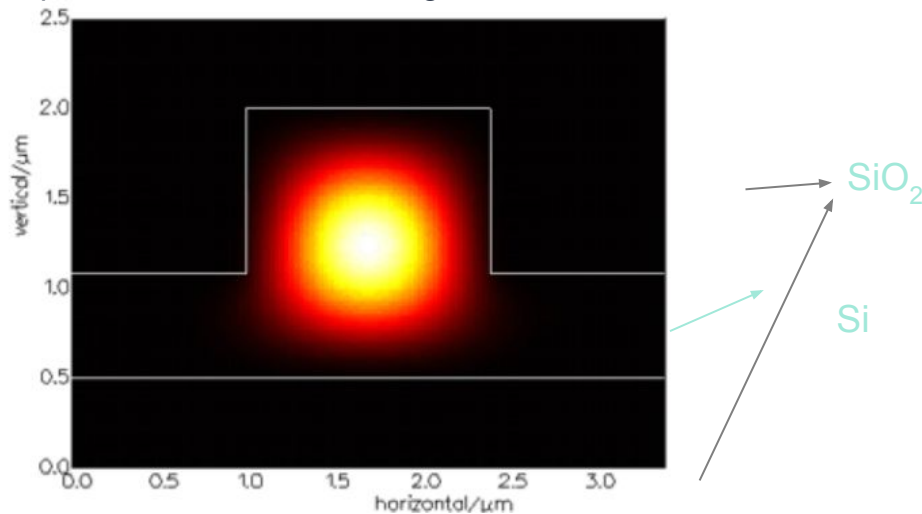
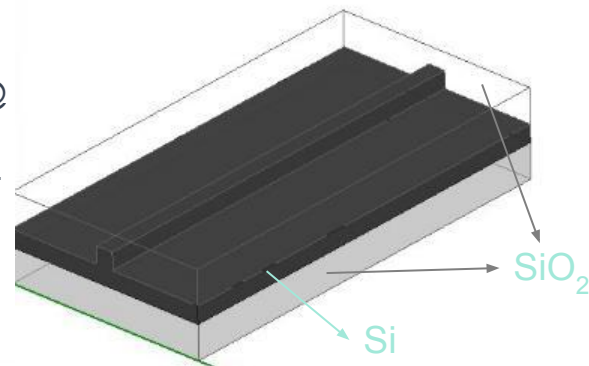


矽光子其中一個關鍵技術是在矽晶圓上製作光波導作為訊號傳輸管道，光波導是利用光在不同折射率邊界會產生全反射的性質，使被侷限在光波導中。

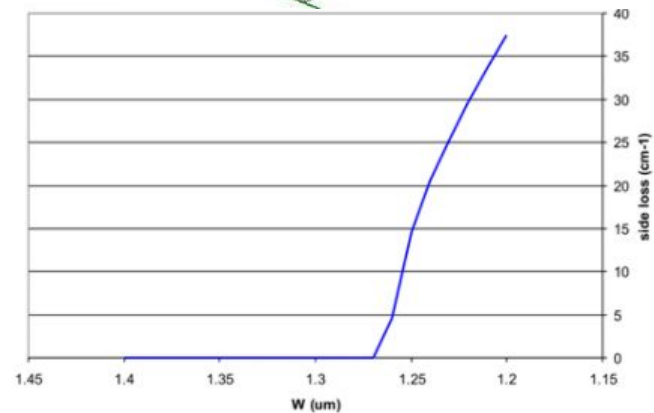
# Waveguide

## SOI wave guide

- The refractive index difference between Si and SiO<sub>2</sub> is significant (3.47 & 1.44 @ 1550nm).
- This high difference of n means it has an excellent ability to confine optical fields.
- The SOI wafer is compatible with the semiconductor process
- The production cost of SOI is high



Good confinement of electric field(rib waveguide)

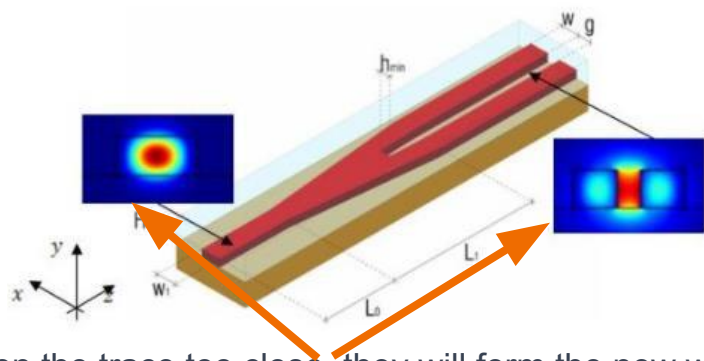


Loss is dependent on wavelength(because of n(reflection index) )

SOI(矽氧化物絕緣體)光波導是一種重要的集成光學平台, 廣泛應用於光通訊領域SOI光波導的核心技術在於利用矽(Si)和氧化矽(SiO<sub>2</sub>)兩種材料的折射率差異來實現光的有效導引。在SOI結構中, 矽層作為導光層, 而氧化矽層則作為底層來支撐矽層並提供光學隔離。SOI光波導在1.3微米( $\mu\text{m}$ )以上的波長表現出低損耗的特性。這一波長範圍不僅涵蓋了光通訊中常用的D與C波段(1.31和1.55微米波段), 這些都是光纖通訊中重要的傳輸窗口。低損耗特性意味著光在傳輸過程中的能量損失極低, 對於提高通訊距離和信號品質至關重要

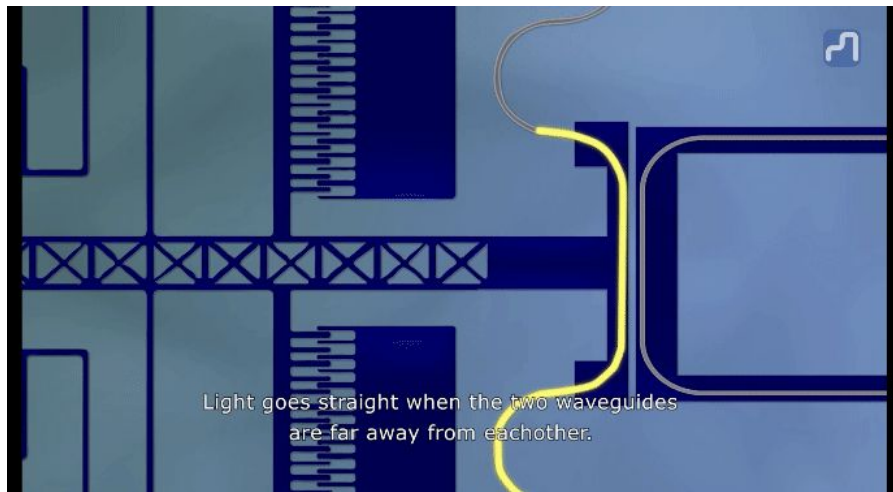
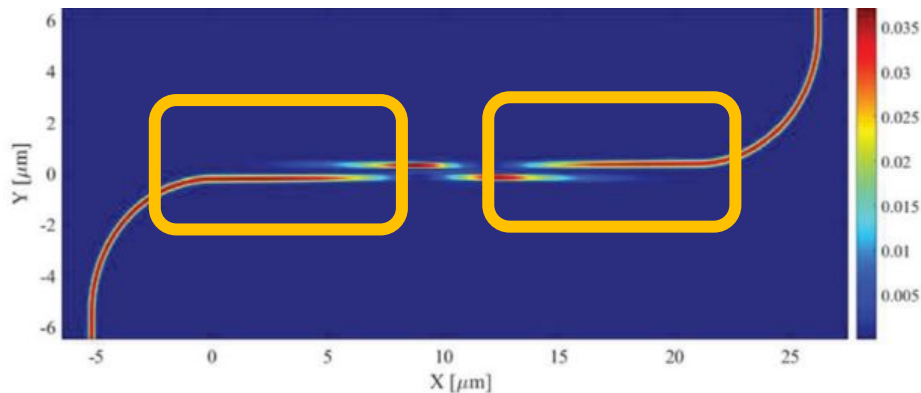
# Directional coupler

Directional coupler principle



When the trace too close, they will form the new wave guide.

Light can couple to another trace under certain conditions.

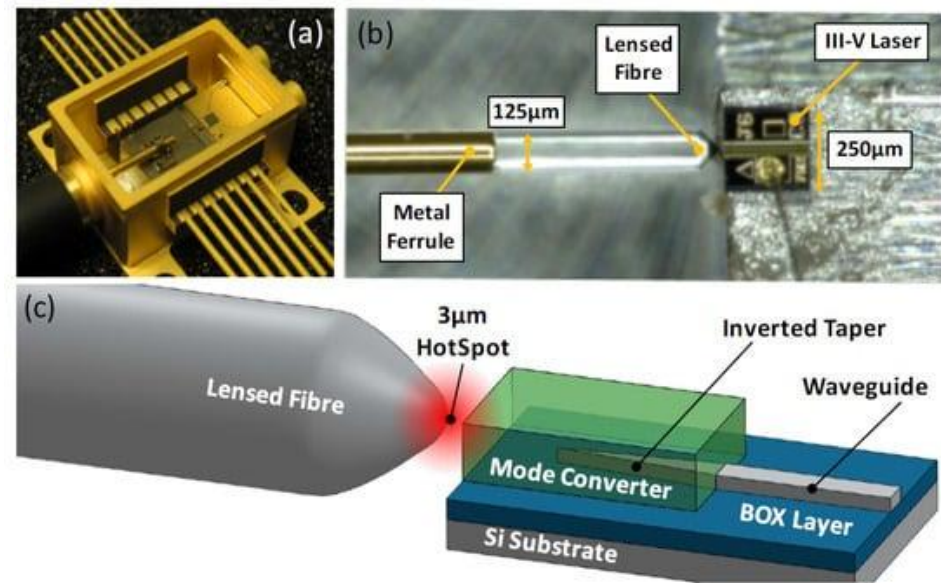


By adjust the distance between trace can control the coupling strength

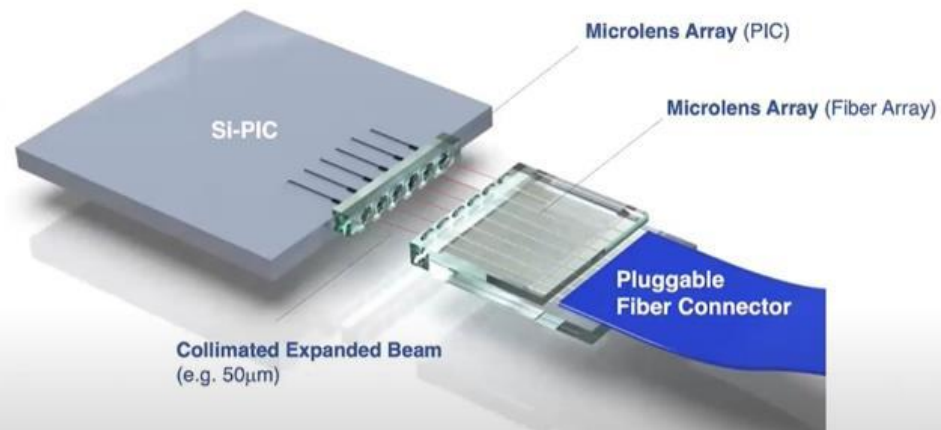
方向耦合器是一種常用於光通訊和集成光學的裝置，能夠將光信號從一條波導耦合到另一條波導。這種耦合過程主要取決於兩條波導間的距離和相互作用長度。

# I/O couple

## Edge couple



## Optical Packaging (micro optics)



邊緣耦合是一種常用於實現高效率光信號從光纖到波導或芯片的轉移的技術。通過使用精確製作的鏡頭光纖和精細設計的模式轉換器，可以最大化入射光信號的耦合效率，從而降低能量損失，提高整體系統的性能。

在光學封裝中，使用微透鏡陣列來聚焦和導引光束至特定位置，這對於矽光子元件（如矽基光集成電路）的精確對接至關重要。這種對接技術支持高密度光路的配置，使得多通道光信號處理成為可能。

# I/O couple

Grating couple

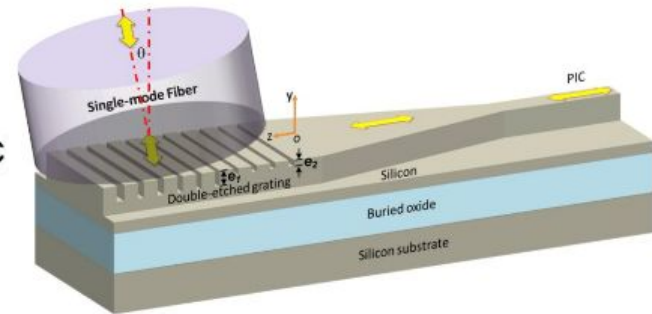
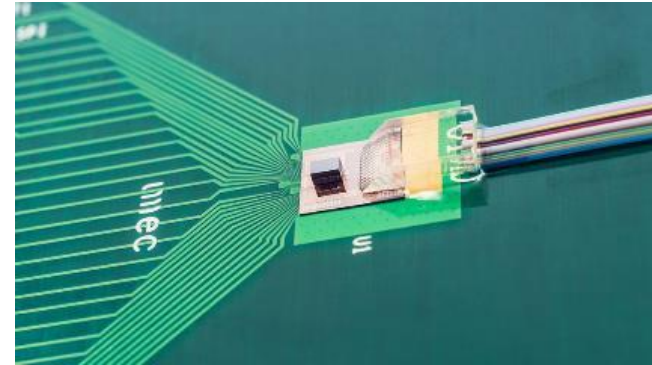
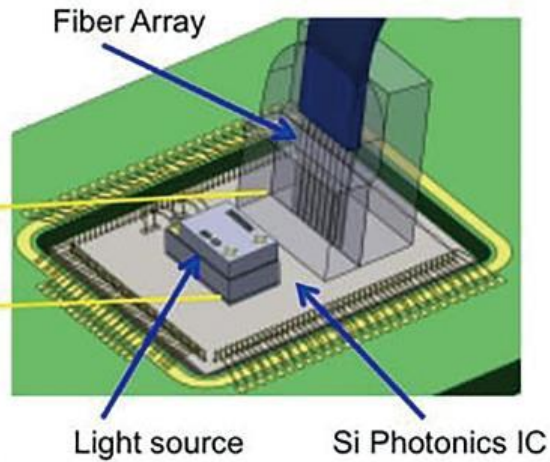
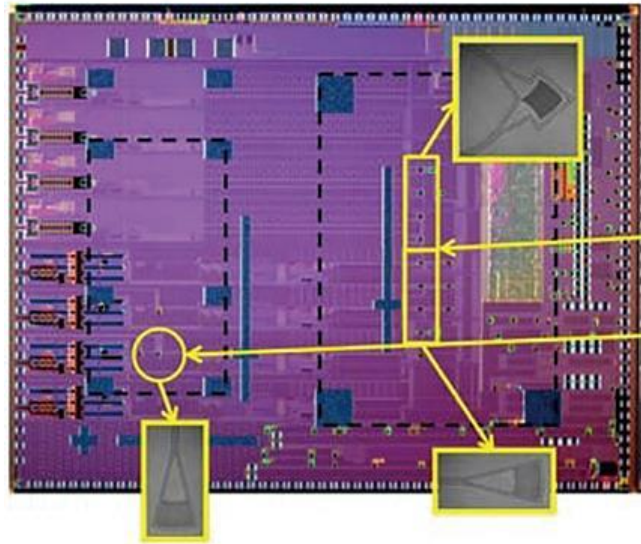
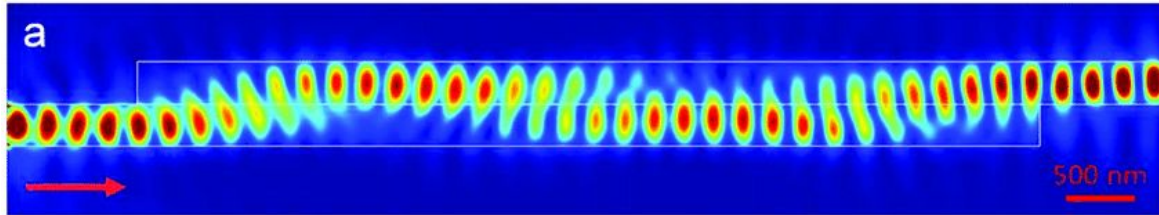
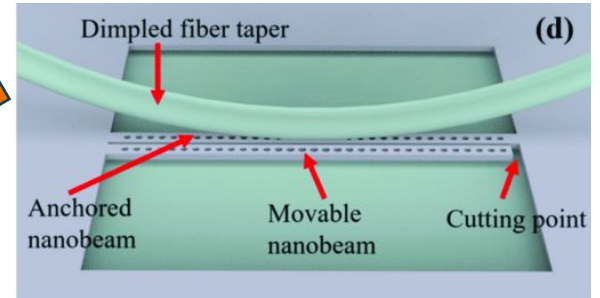
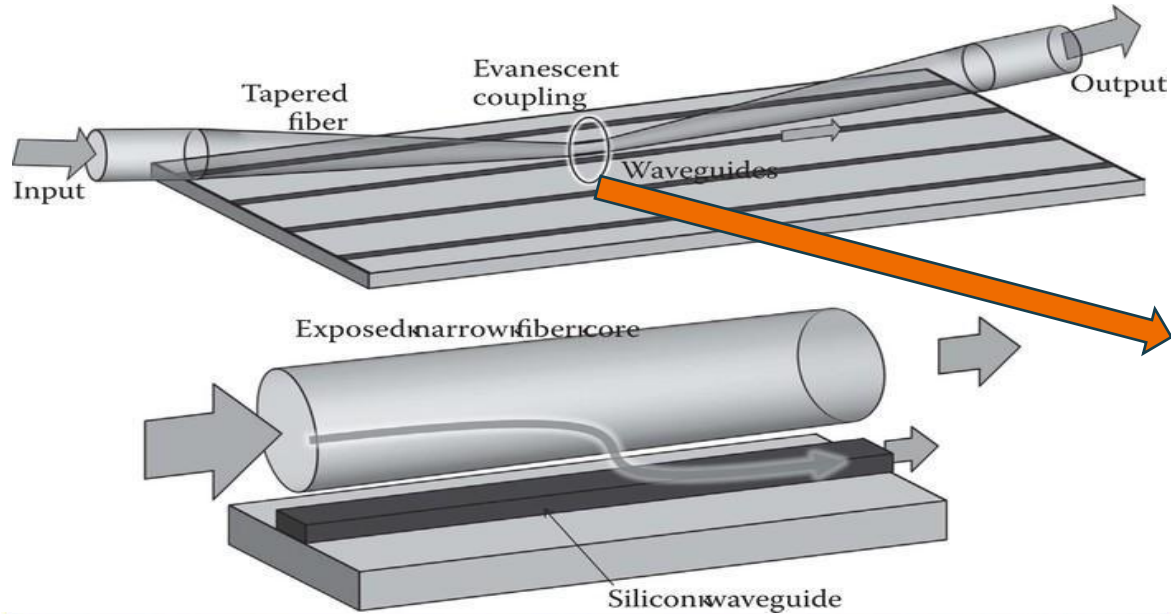


Fig. 1. Schematic of the double-etched diffractive apodized waveguide grating coupler.

# I/O couple

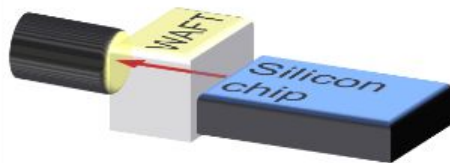
## Evanescent-wave coupling



# I/O couple

## Waveguide Array to Fiber Transposer

Edge coupling

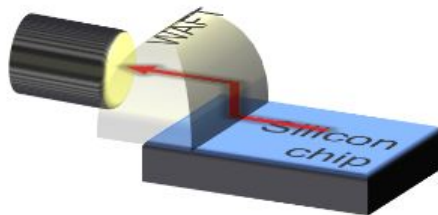


Low loss solution  
Compatible with WDM



Wafer level incompatible  
Alignment accuracy

Top coupling

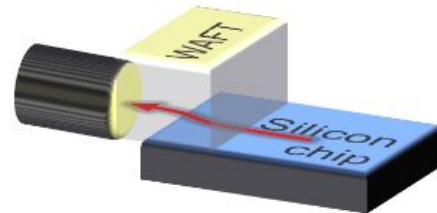


Compatible with grating  
couplers  
Wafer level compatibility



Grating coupler loss  
WDM-limited

Evanescent coupling



Low loss solution  
Compatible with WDM  
Alignment tolerant  
Wafer level compatibility

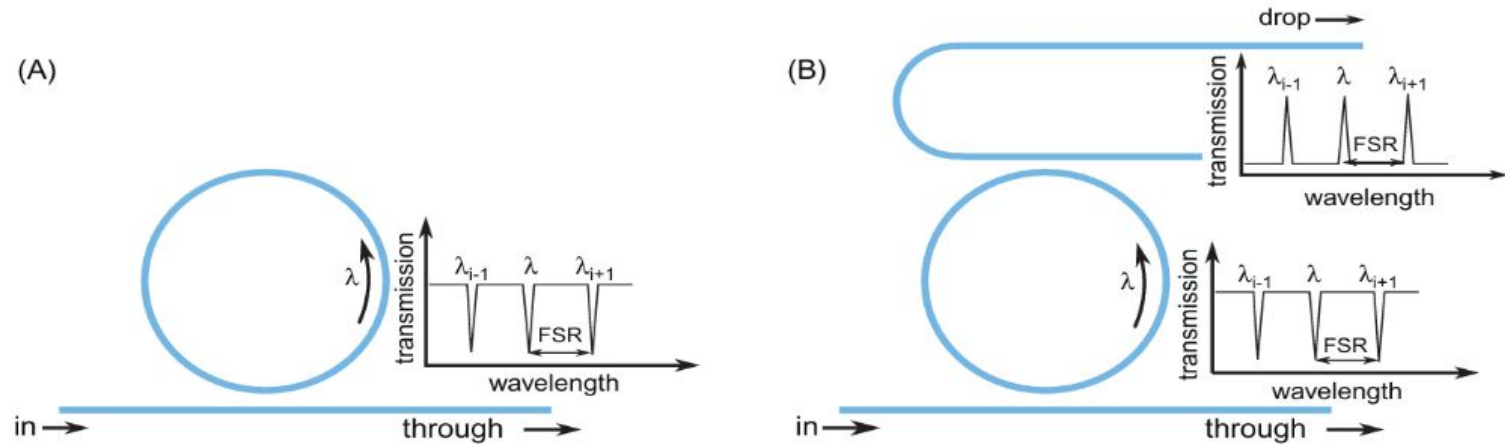


BEOL constraint

Chips will usually have packaged connectors.  
One issue to consider here is whether it is WDM or single  
frequency.

# Ring resonator

## Principle of ring resonator



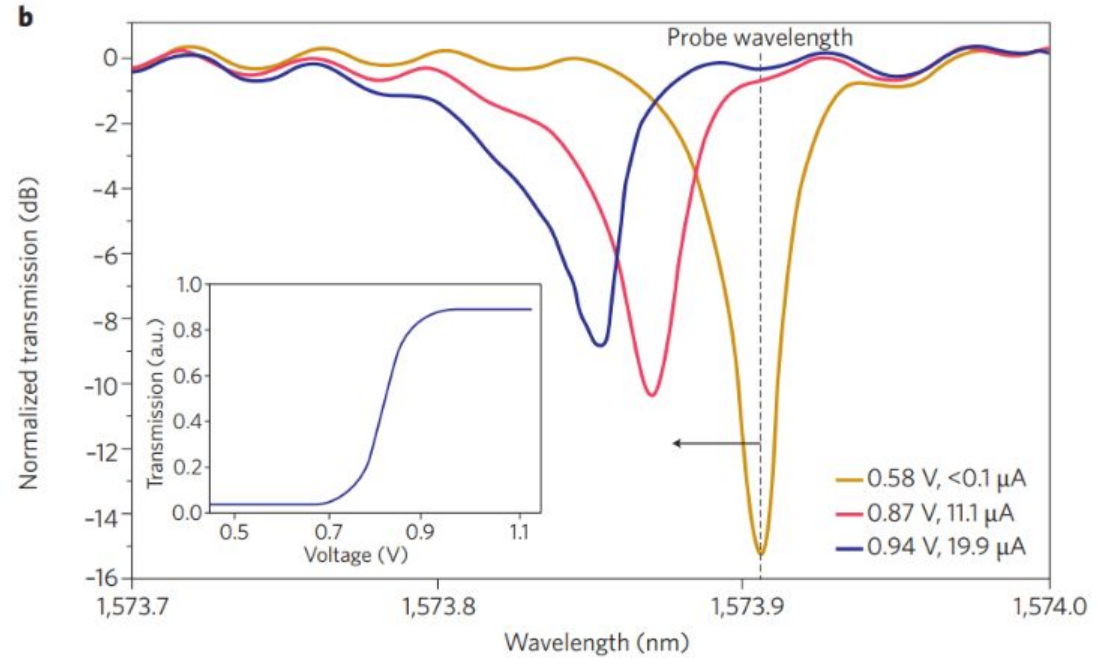
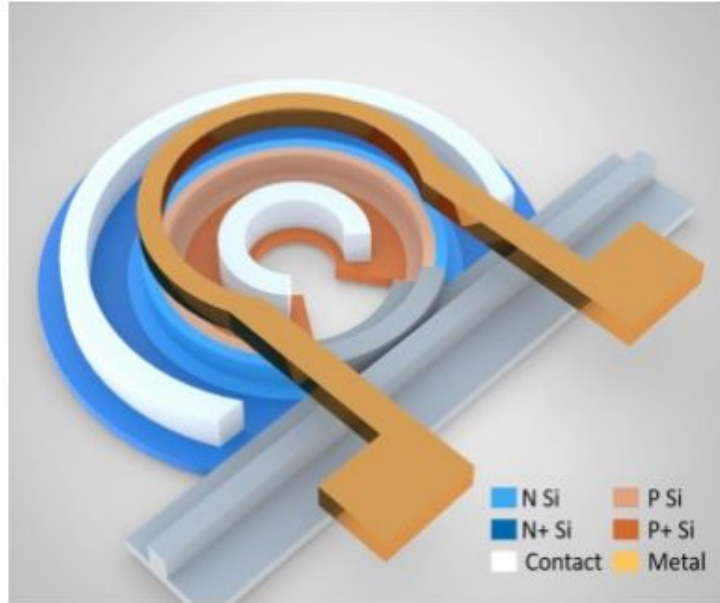
**Figure 7.20**

(A) Working principle of a ring resonator with single bus waveguide. (B) Working principle of a ring resonator with double bus waveguide.

1. A portion of the light **enters the ring** structure, while the rest continues to propagate forward.
2. Subsequently, these light waves interfere again in the coupling region.
3. Destructive interference occurs when their phase difference reaches  $\pi/2$ .
4. This type of resonator has a **simple structure**, compact size. Its high Q factor indicates low energy loss, but conversely, it has a very **narrow bandwidth**.

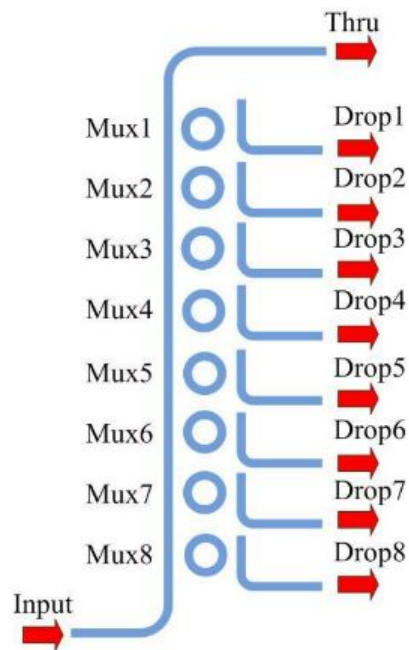
# Micro-Ring Modulator

Use the ring resonator and EO(Plasma Dispersion effect) effect or TO(Thermal) to make modulator



# Ring cavity

Using as DEMUX or MUX



3-7 利用塞取環狀共振器製作之高密度分波多工器

**LET'S MAKE A TRANSCEIVER**

Example: A 100Gbps transceiver

PSM4: 4 parallel fibers

QAM16: coherent

WDM4: wavelength channels

4 MZI modulators

wavelength multiplexer

Tx

wavelength demultiplexer

Rx

4 lasers in

4 photodetectors

fiber out

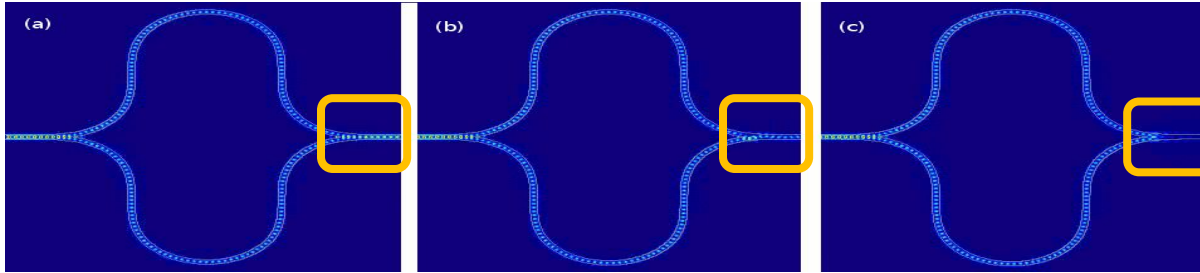
fiber

GHENT UNIVERSITY

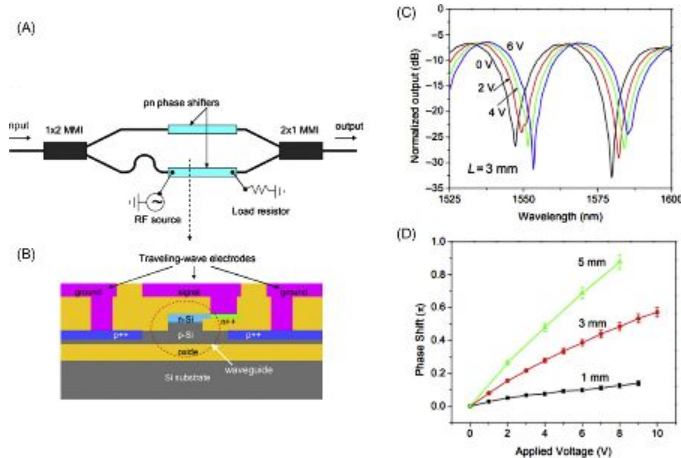
# Mach-Zehnder modulator

## Principle of MZM

Following is a demonstration of a simple Mach-Zehnder Interferometer (MZI) with phase shifts of 0,  $\pi/4$ , and  $\pi/2$



0 : constructive interference,  $\pi/4$  : linear control range,  $\pi/2$  : destructive



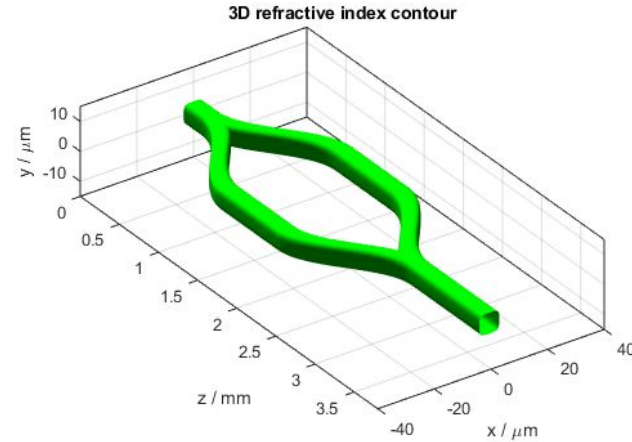
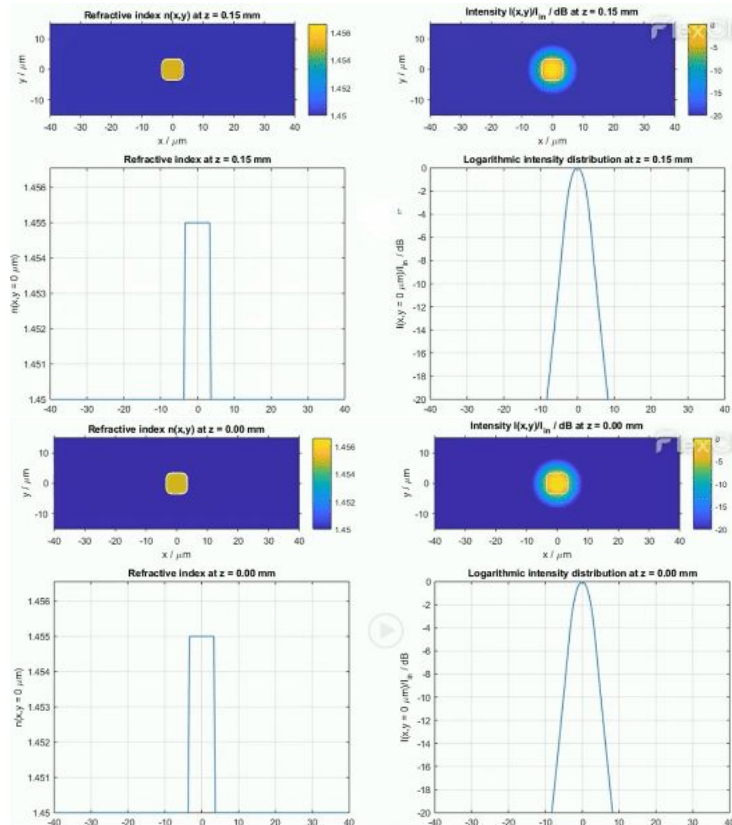
- Left is a standard Mach-Zehnder modulator that incorporates both a basic MZI and a phase modulator.
- PN semiconductors are used as the modulating elements.
- By adjusting the carrier concentration through voltage, the refractive index is changed. (plasma dispersion effect)
- Also, can control the phase difference between two separate optical paths.

<https://medium.com/lightmatter/optical-components-for-matrix-processing-73687cdd7aa6>

High-performance Mach-Zehnder modulator using tailored plasma dispersion effects in an ITO/ graphene-based waveguide

# Mach-Zehnder modulator

Principle of MZM for x,y plane

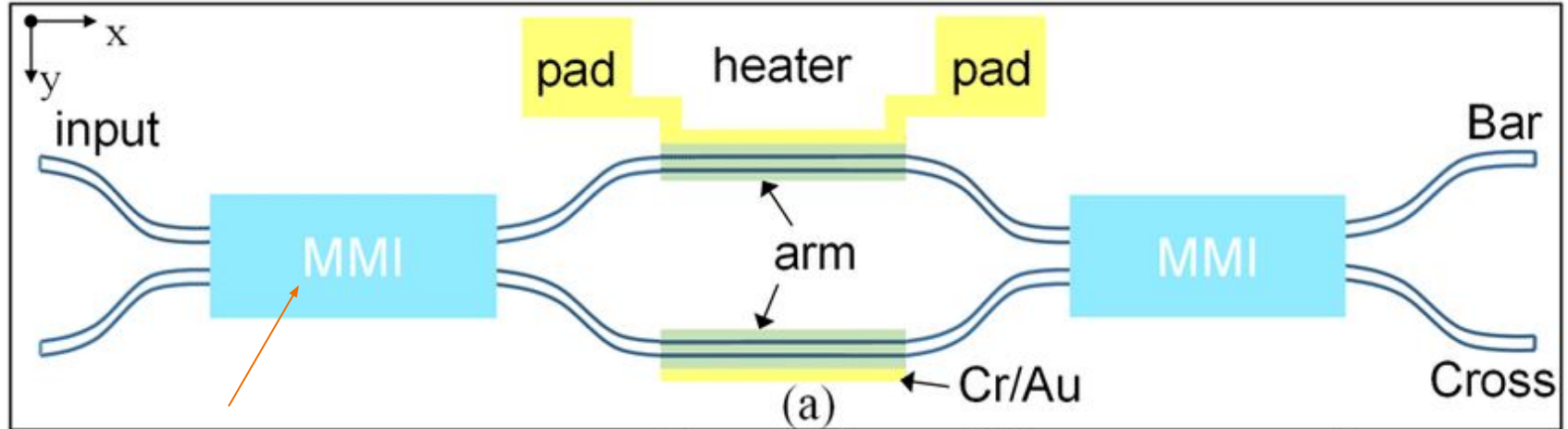


1. Initially, light is split through a -3dB directional coupler.
2. Subsequently, the light travels through two distinct paths.
3. When we adjust the refractive index of one of the paths, a phase difference occurs.
4. The light are combined again through a -3dB directional coupler.
5. They interfere with each other.
6. Destructive interference occurs when the phase difference reaches  $\pi/2$ .

# Mach-Zehnder modulator

Thermo-optic effect

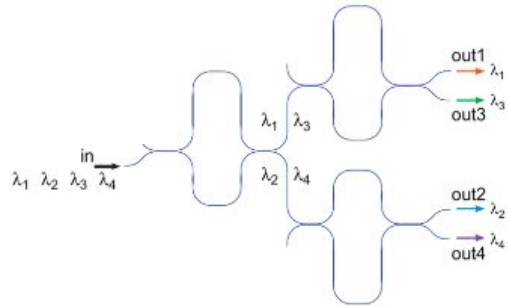
- We can also use the thermo-optic effect for phase modulation, but its speed is much slower(ms us) compared to the electro-optic effect.



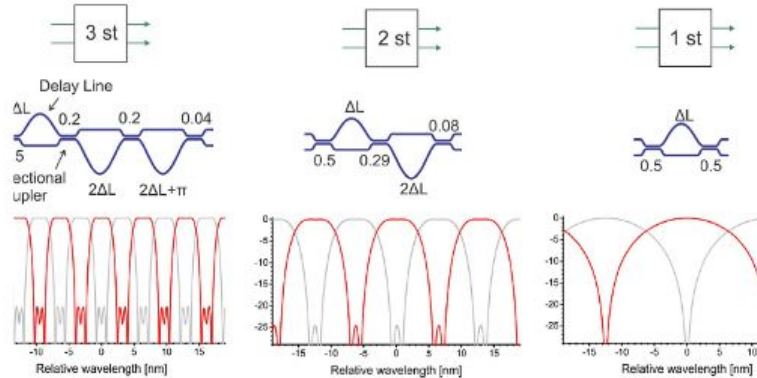
Multi-Mode Interference (MMI)  
Coupler

# Mach-Zehnder modulator

Other MZM example :



Use MZM as the demux



High order  
MZM

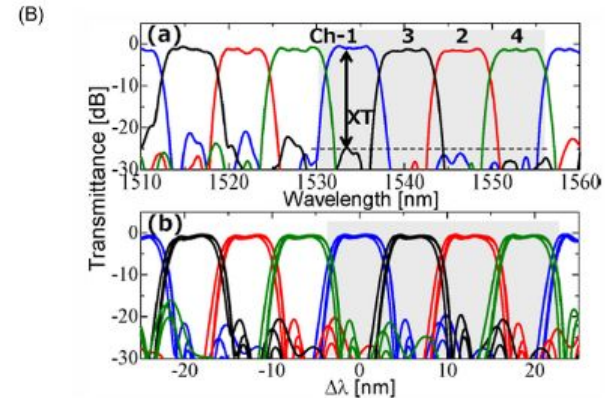
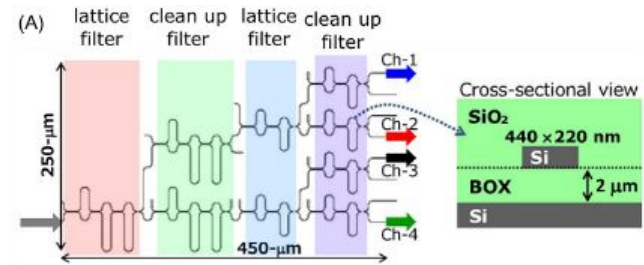
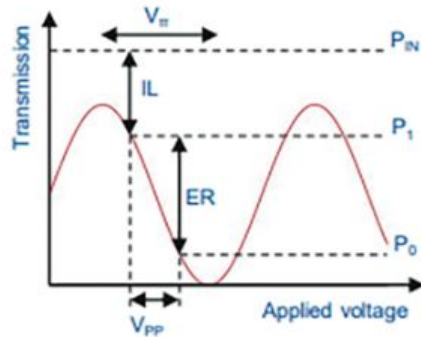
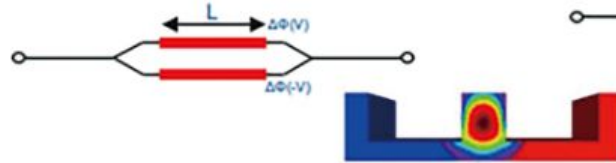


Figure 7.19

(A) Silicon four-channel 800 GHz lattice filter design. (B) Measured spectral response four-channel 800 GHz lattice filter. From [23] S.H. Jeong, Y. Tanaka, K. Morito, *Silicon photonics optical demultiplexer technology for wdm optical interconnects*, in: 2015 Opto-Electronics and Communications Conference (OECC), 2015, pp. 1–3.

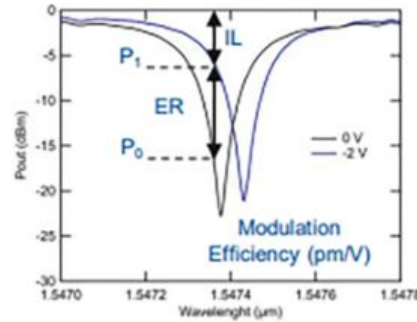
# MZM vs. MRM

Si Mach Zehnder Modulator



- ☐ Optically broadband
- ☐ Thermally robust
- ☒ Large footprint ( $L \sim 1\text{mm}$ )
- ☒ High driving voltage amplitude

Si Ring Modulator

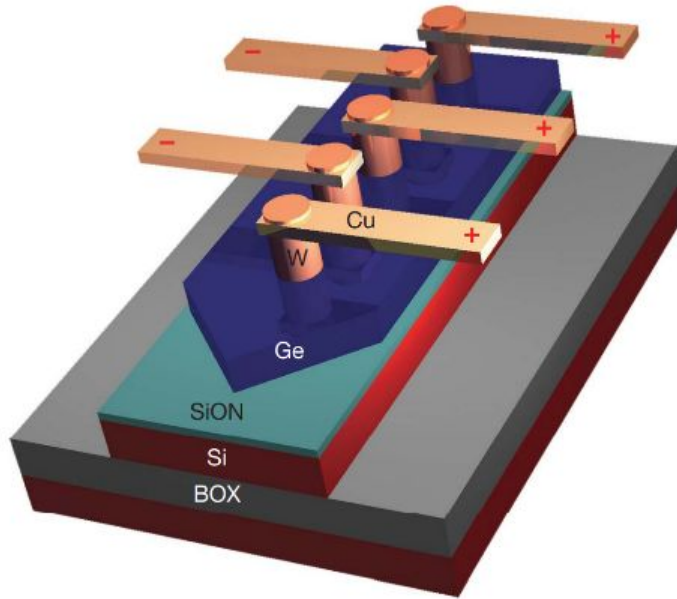


- ☒ Optically narrowband ( $< 1\text{nm}$ )
- ☒ Thermally sensitive ( $< 1\text{K}$ )
- ☐ Small footprint ( $R \sim 5\mu\text{m}$ )
- ☐ Small driving voltage amplitude ( $1\text{Vpp}$ )

- The frequency response of MZM is broader and flatter.
- MRM has a narrower bandwidth, and its peak variation is sharper, but it requires a smaller footprint
- We usually mix MZM and MRM in design to get the broadband and small footprint.

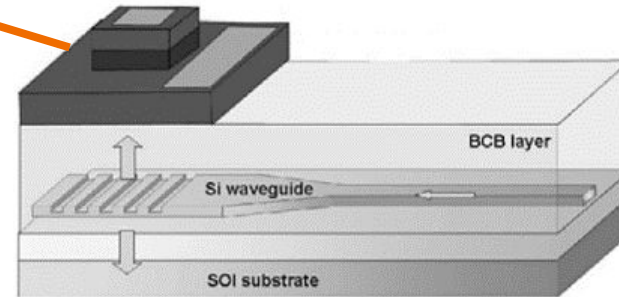
# Ge on Si Photonics

- The operational bandwidth of the silicon waveguide heterogeneously integrated germanium photodiode can reach above 35GHz.

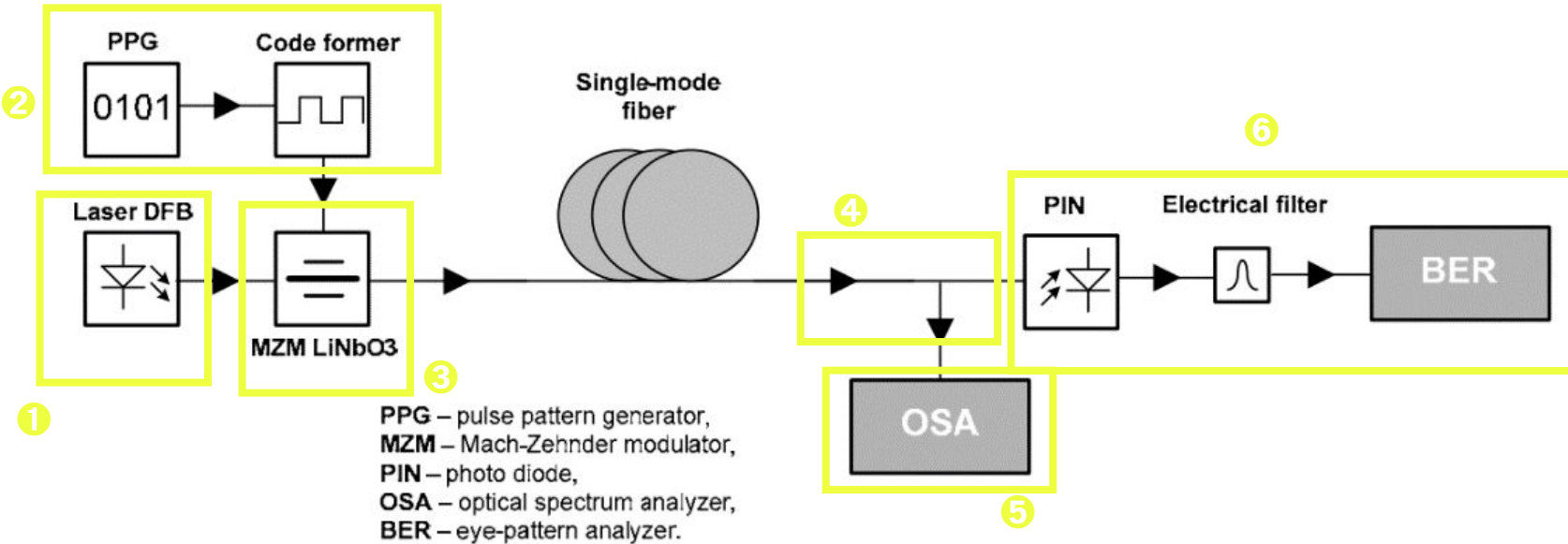


## 100 Gbps RZ Data Reception in 67 GHz Si-contacted Germanium Waveguide p-i-n Photodetectors

H. Chen, M. Galili, P. Verheyen, P. De Heyn, G. Lepage, J. De Coster, S. Balakrishnan, P. Absil, L. Oxenlowe, J. Van Campenhout, G. Roelkens



# Simplified optical transmission system

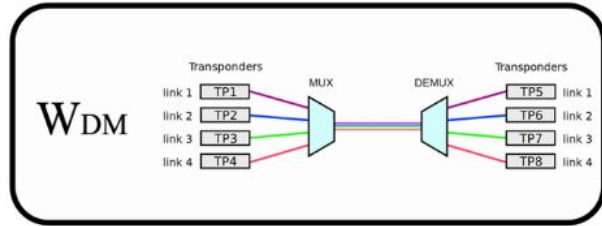


The diagram outlines a basic optical communication system:

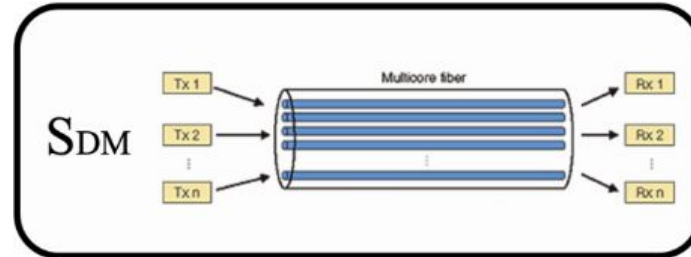
1. A continuous light source (typically a laser) is essential.
2. Electronic signals are encoded into an amplitude modulator (either EOM or EAM).
3. This modulated light conveys information to our DUT.
4. A directional coupler divides the signal between the OSA and the high-speed O2E (high speed photodetector)
5. Use OSA to monitor the spectrum
6. The O2E converts the optical signal into electronic and see its signal quality.

# Protocol to enhance the data capacity

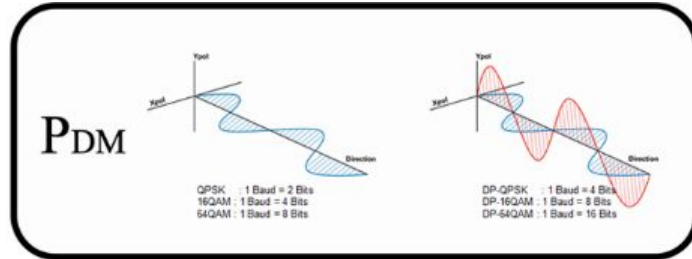
WDM, SDM, MDM, PDM



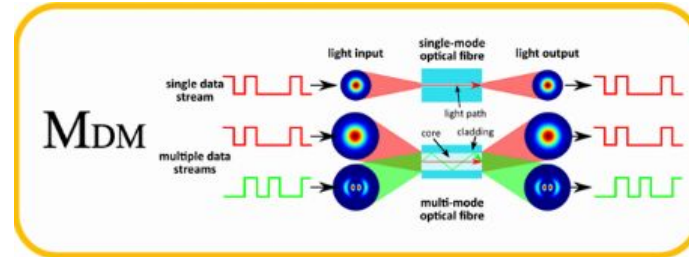
WDM(Wavelength-division multiplexing)



SDM(space-division multiplexing)



PDM(Polarization-division multiplexing)

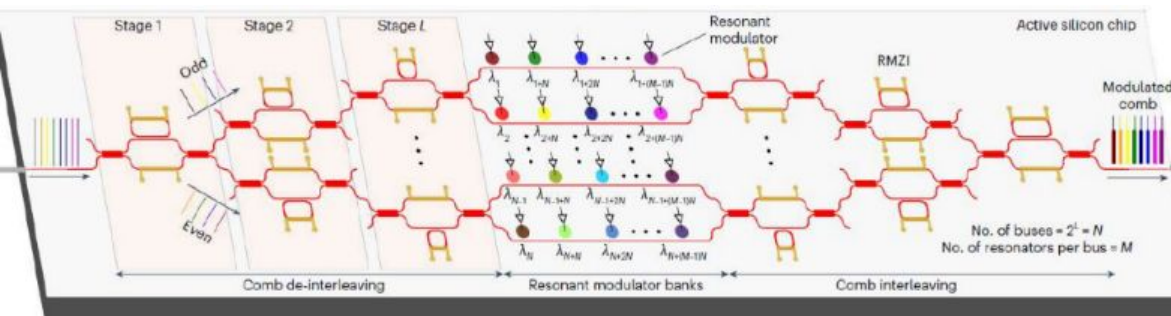


MDM(Mode Division Multiplexing)

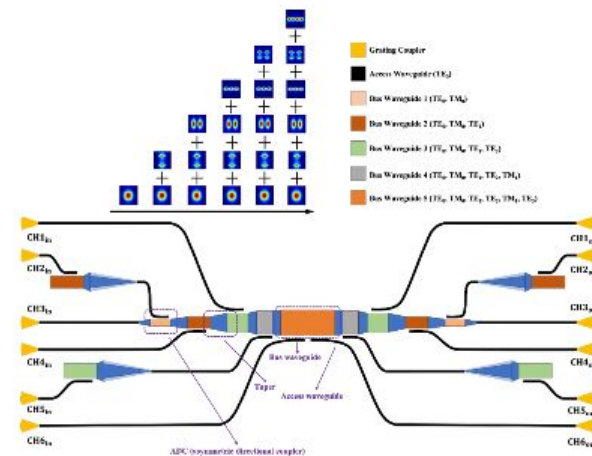
Other methods:

Advanced signal modulation techniques such as Pulse Amplitude Modulation (PAM) or Coherent are gradually being adopted to enhance the overall transmission capacity of optical transceivers.

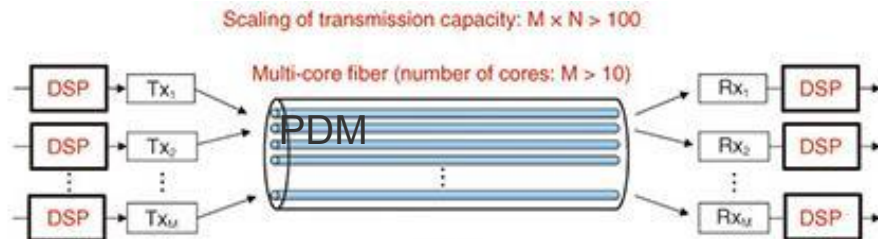
# Example of WDM, SDM, MDM, PDM



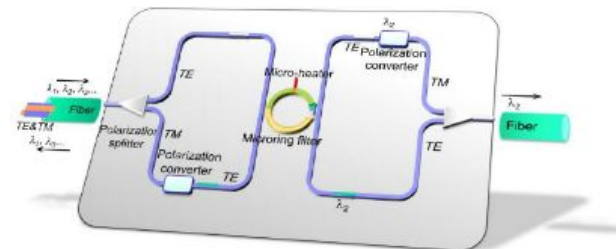
WDM



MDM



SDM



PDM

# Example of WDM, SDM, MDM, PDM

Combining various XDM is also a potential direction

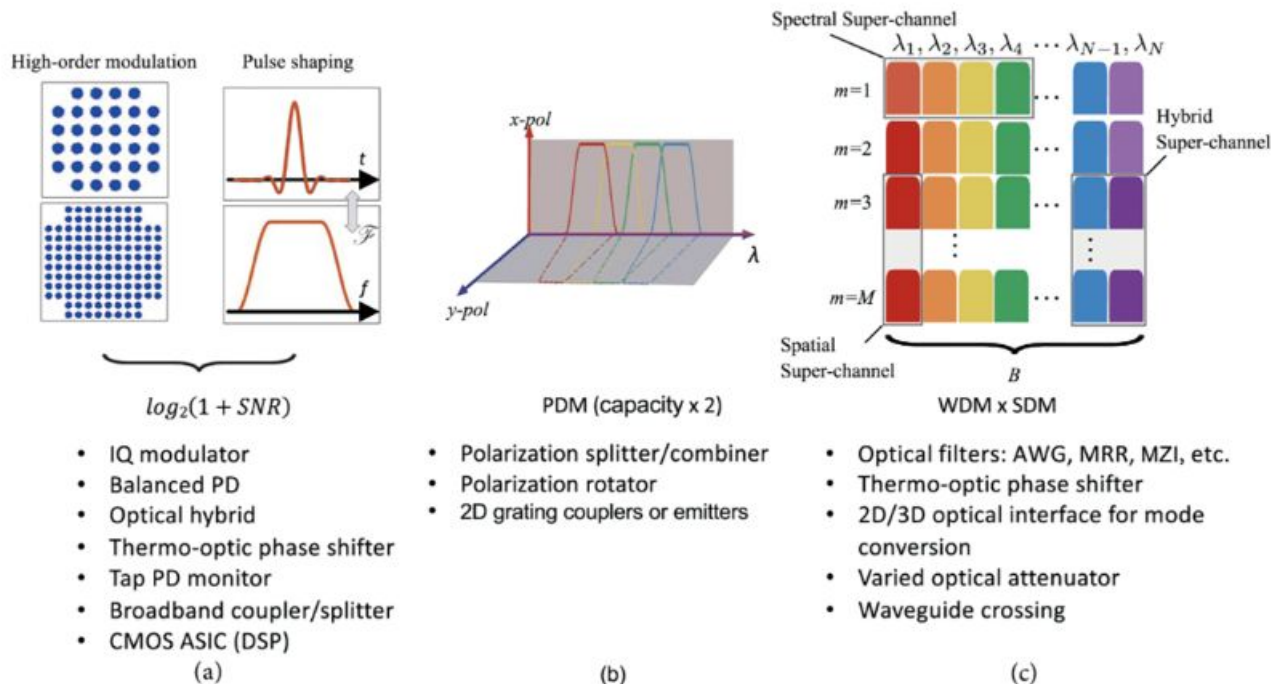
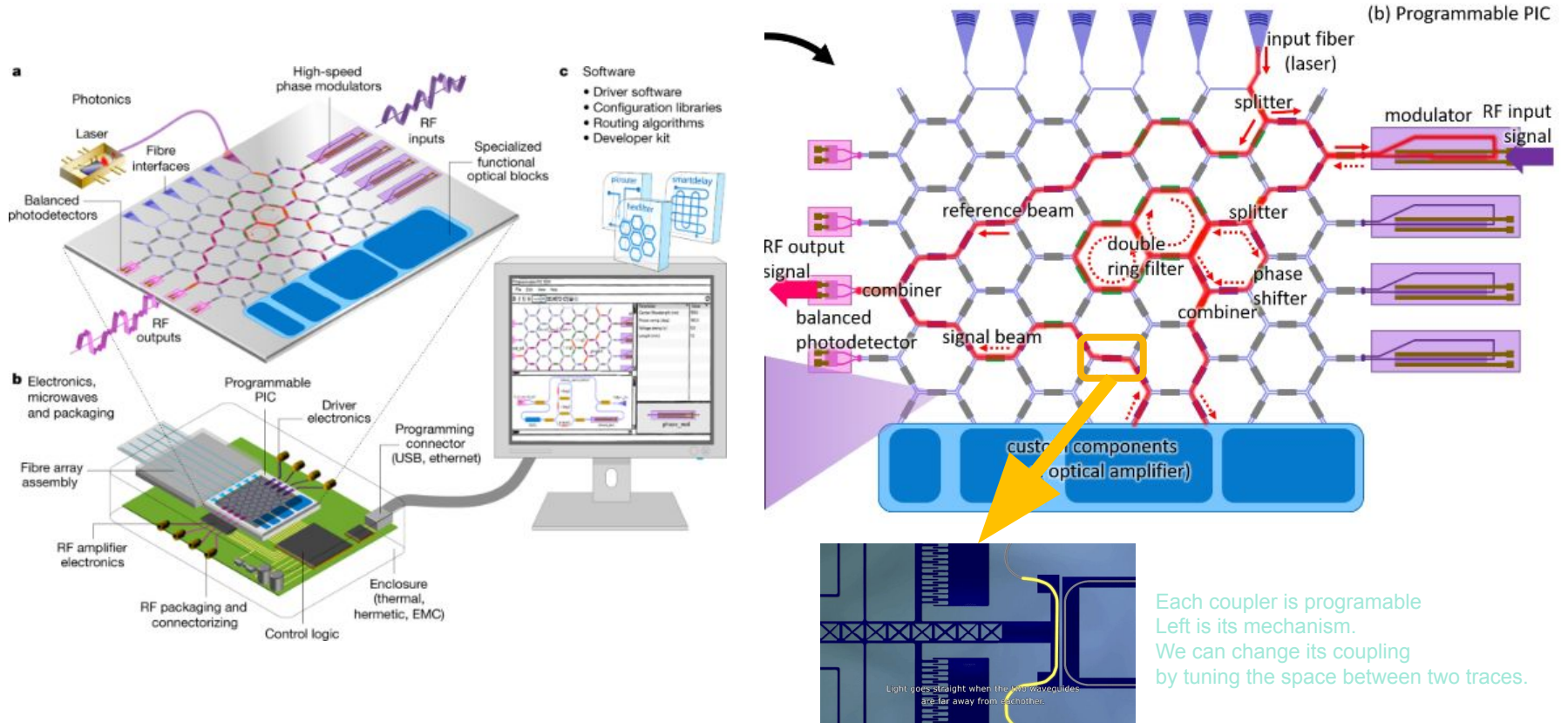


圖2 各種提高整體傳輸速率的方法及所需的元件，(a)光同調傳輸，(b)極化多工傳輸，(c)空間分割多工與分波多工的整合傳輸 (Ref. Wei Shi, et al. Nanophotonics)。

# Programmable PIC

By a lot of directional coupler we can make a lot of logical unit



# 台廠矽光子/CPO供應鏈

| 主要供應    | 公 司 (代號)                                               |
|---------|--------------------------------------------------------|
| 磊晶製造    | 全新 (2455)、聯亞 (3081)                                    |
| 晶圓設計與代工 | 台積電 (2330)                                             |
| 光收發模組   | 上詮 (3363)、波若威 (3163)、華星光 (4979)、訊芯-KY (6451)、前鼎 (4908) |
| 交 換 器   | 智邦 (2345)、明泰 (3380)                                    |
| 封裝測試    | 日月光 (3711)、矽格 (6257)、台星科 (3265)                        |

資料來源：採訪整理

製表：李娟萍、張珈睿

■ 矽光計劃產業供應鏈



# Waveguide

## SWG waveguide

- SWG waveguides have a periodicity below the Bragg condition
- Can adjust refractive index and dispersion without altering fabrication thickness.
- Used for various optical functions like polarization control and wavelength filtering.
- Uses Bloch mode for low-loss light transmission.



圖2-6 一般直波導示意圖

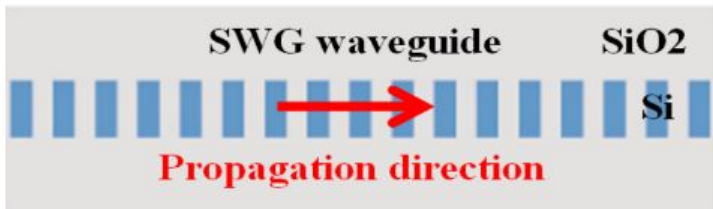


圖2-7 次波長光柵光學波導示意圖

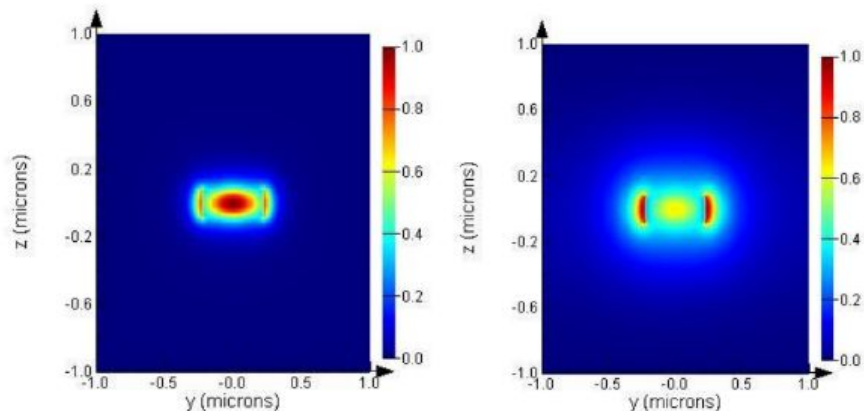


圖2-9 (a)一般直光學波導模態分布 (b)次波長光柵波導模態分布

| 產業      | 股票代碼 | 股票名稱  | 股價 (截至<br>2024/04/26 ) |
|---------|------|-------|------------------------|
| 磊晶      | 3081 | 聯亞    | 123                    |
|         | 2455 | 全新    | 142.5                  |
| 晶片設計與代工 | 2330 | 台積電   | 785                    |
| 光收發模組   | 3163 | 波若威   | 100                    |
|         | 4979 | 華星光   | 115.5                  |
|         | 4977 | 眾達-KY | 72.8                   |
|         | 3234 | 光環    | 44.95                  |
|         | 6442 | 光聖    | 134                    |
|         | 3363 | 上詮    | 121.5                  |
|         | 6451 | 訊芯-KY | 171                    |
|         | 4908 | 前鼎    | 100.5                  |
| 交換器     | 2345 | 智邦    | 419.5                  |
|         | 3380 | 明泰    | 32.35                  |
| 封測      | 3711 | 日月光   | 143.5                  |
|         | 6257 | 矽格    | 71.2                   |
|         | 3265 | 台星科   | 125.5                  |
| 測試介面    | 6223 | 旺矽    | 409                    |
|         | 6515 | 顯崙    | 857                    |

[https://www.pida.org.tw/main/article\\_show/0279](https://www.pida.org.tw/main/article_show/0279)

<https://www.degruyter.com/document/doi/10.1515/nanoph-2020-0309/html>